

Evolution and Biology for Artificial Intelligence

Course Syllabus

Zeki Doruk Erden

Course Schedule

Note: Since this is the first offering of the course, the schedule may be subject to minor adjustments during the semester as needed.

Part 1: Introduction to Biology and State of Technology in Bio-Inspired AI

- **Lecture 1:** Introduction and overview
- **Lecture 2:** Artificial intelligence, past and present
- **Lecture 3:** Neural systems
- **Lecture 4:** Classical evolutionary theory
- **Lecture 5:** Evolutionary computation
- **Lecture 6:** Developmental systems
- **Lecture 7:** Collective systems
- **Lecture 8:** Artificial life

Part 2: Extended Evolutionary Theory and the Prospects for the Future of AI

- **Lecture 9:** Evolution & intelligence relation
- **Lecture 10:** Principles of biological organization & patterns of evolutionary change
- **Lecture 11:** Evolutionary developmental biology
- **Lecture 12:** Major transitions in evolution
- **Lecture 13:** A discussion on evolutionary theory, AI and singularity

Subject to time availability, the concluding one or two weeks of the course may be devoted to Q&A sessions and project proposal presentations.

Resources

The following books act as references and further reading suggestions for the corresponding lectures listed alongside them. Additional papers mentioned during lectures may also be consulted at the relevant points for deeper exploration.

- *Bio-Inspired Artificial Intelligence: Theories, Methods, and Technologies* – Floreano & Mattiussi [L3 to L8]
- *Neuroevolution: Harnessing Creativity in AI Model Design* - Risi et al. [L5]

- *Swarm Intelligence: From Natural to Artificial Systems* – Bonabeau, Dorigo, Theraulaz [L7]
- *Evolution and the Theory of Games* – Maynard Smith [L7]
- *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior* – Friston, Parr, Pezzulo [L9]
- *Endless Forms Most Beautiful* – Carroll [L10, L11]
- *Plausibility of Life* – Kirschner & Gerhart [L10, L11]
- *Developmental Plasticity and Evolution* – West-Eberhart [L10, L11]
- *Major Transitions in Evolution* – Maynard Smith & Szathmáry [L10, L12]
- *On the Origin of Autonomy* – Rosslenbroich [L12]

Evaluation

30% Project Proposals

You will prepare two *project proposals* (each worth 15%) during the semester: one based on a pre-set project topic and one on a free-choice topic.

Each proposal should include the problem definition, its significance, relevant background information, and a detailed description of your proposed methodology (system design, algorithmic flows, test setups, etc.) for studying and addressing the problem. *Since this is not a coding-oriented course, you are not expected to implement the system, run experiments, or perform any results analysis.* Each proposal should be submitted as a report of at most 6 pages, including all figures and tables, excluding references.

Pre-set project topics (details will be provided in a few weeks):

- Project 1: Implementing a Growing Neural Network (suggested after Lecture 6)
- Project 2: Swarm Behavior for Harvesting (suggested after Lecture 7)

These projects provide the research or engineering question but allow you are open-ended in the methodology you may choose to follow.

For the free-choice project, you may select any topic relevant to the course that poses a meaningful conceptual and technical challenge and is suitable as a scientific or engineering question. It must be distinct from, and not too similar to, the pre-set project topics. (If in doubt about your topic, consult me verbally or by email for approval.) This project will also include a short presentation based on your proposal report.

Projects may be completed individually or in groups of 2. Because of the large class size, group work is heavily encouraged; individual submissions will not receive a positive bias in grading.

Deadlines for reports will be announced later.

70% Exams

There will be one midterm exam (30%) and one final exam (40%). Exams will be theoretical in nature and biased towards (but not exclusively on) the evolution & biology content of the course.